rsna-knee-abnormality-detection

2nd Place Solution

Rank	2
Team	MURAD
Author	azamat1ch
Topic ID	752370
Version	Original

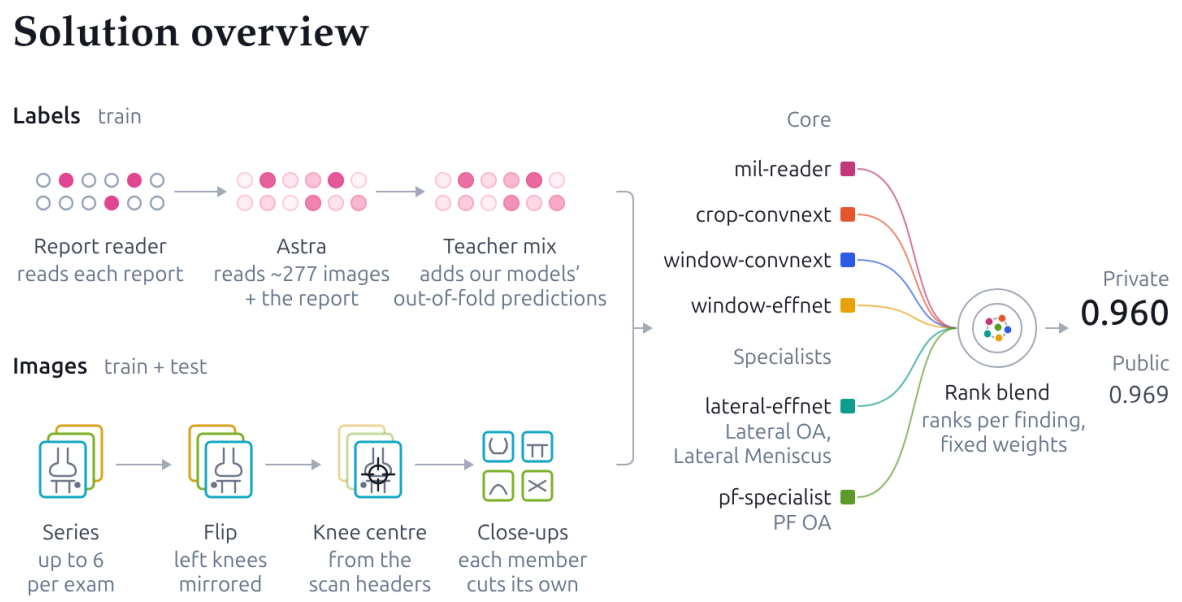
Source: https://www.kaggle.com/competitions/rsna-knee-abnormality-detection/discussion/752370
Retrieved with Kaggle CLI on 2026-10-27.

2nd place. Private 0.960, public 0.969.

Our sincere thanks to RSNA for building this dataset, to the radiologists who graded every test exam from the images, and to Kaggle for hosting. This was our first serious competition, and a stressful one: submissions that crashed on the hidden test, a long stretch of 0.001 steps, and a **final (winning) submission** sent in the last slot of the last day.

1. Solution overview

- **Labels.** An LLM read each radiology report, a second LLM (Astra) read every training exam's images with the report, and our own models' predictions were mixed in last, all as soft labels. Scored against 58 expert-labelled exams, the labels went from 0.899 (reports only) to 0.956.
- **Models.** Six 2.5D MIL models, each cutting its own close-ups of the knee and pooling them into one score per finding.
- **Ensemble.** A rank blend with weights set by hand. Four core members vote on all 12 findings; two specialists add a 0.20 share on PF OA and on Lateral Meniscus and Lateral OA.



Top: how the training labels are made, once. Bottom: what every exam goes through, in training and at test.

Member	Backbone	Votes on	Alone, public / private	Blend weight
mil-reader	ConvNeXt-T, fastMRI-pretrained	all 12	0.966 / 0.957	0.35
crop-convnext	ConvNeXt-T	all 12	0.964 / 0.955	0.25
window-convnext	ConvNeXt-T	all 12	0.964 / 0.955	0.20
window-effnet	EfficientNetV2-S	all 12	0.965 / 0.954	0.20

Member	Backbone	Votes on	Alone, public / private	Blend weight
lateral-effnet	EfficientNetV2-S	Lateral Meniscus, Lateral OA	Lateral OA probe: 0.932 / about 0.944	0.20
pf-specialist	ConvNeXt-T, fastMRI-pretrained	PF OA	PF OA probe: 0.944 / about 0.932	0.20

2. The problem

Training has reports only; the test is graded by experts reading the images. On the 58 training exams with expert labels ("the 58"), the experts mark 240 positives where the reports state 154 (PF OA: 21 against 2), mostly mild disease a report has no reason to mention. The real task was telling mild from normal, and the reports could not teach it.

3. Labels

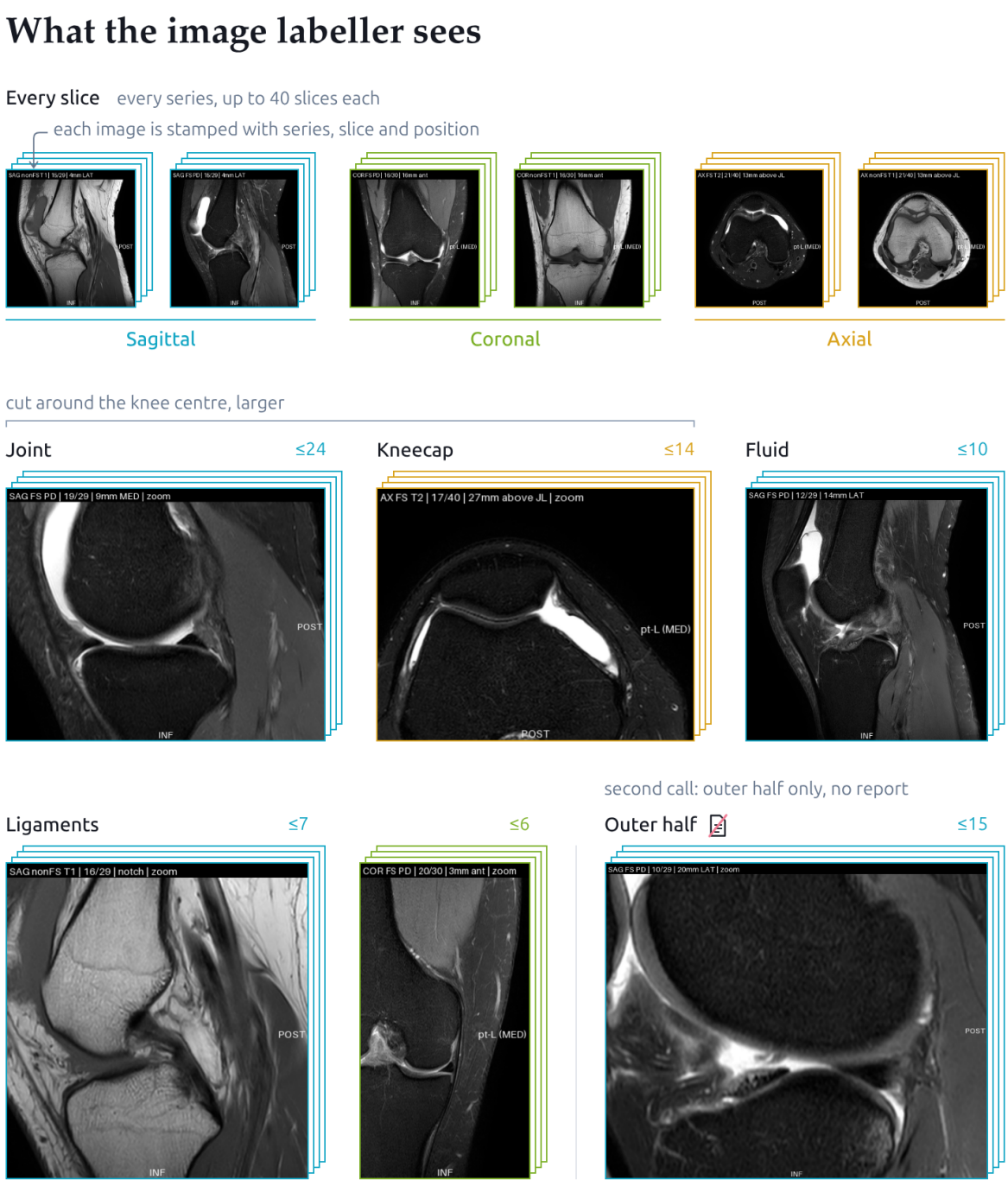
Only training exams were labelled; nothing at inference uses an LLM (the host allowed commercial LLM APIs, thread 733965). We scored every label version on the 58 like a model. A score there moves by about ±0.02 by chance, so we tuned on 24 and kept 34 to check each frozen version; later rules, such as the PF OA floor, used all 58, so those gains are clues, not proof.

3.1 Reading the reports

GPT-6.1 Sol read each report in its own language. Each language is roughly one hospital, so a translation slip would teach the model the hospital, not the knee. For each finding it quotes the report, picks one of six answers (yes, probably yes, unclear, probably no, no, not mentioned) and gives a 0-100 score. Code turns the answer and score into a soft label. These labels scored 0.899 on the 58.

3.2 Reading the images

An early pilot said images would not help, but it had shown the model only 44 of 134 slices. So Astra (GPT-6 Astra) got every slice: one call per training exam, about 277 images, with the report and the reader's answers. It was told to treat the report as a colleague's read with known gaps.



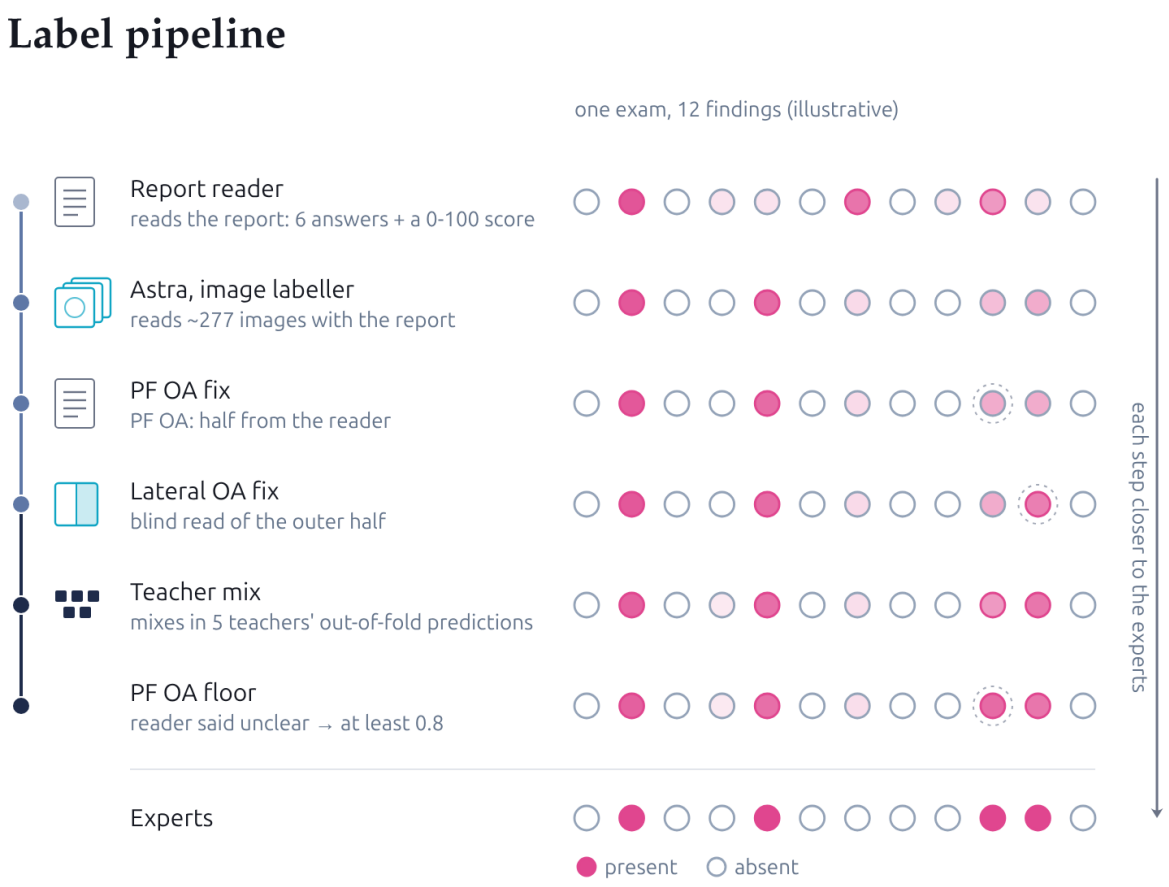
Every series first, then larger close-ups; the number above a close-up stack is the most slices it holds.

Astra's labels scored 0.942 against the reader's 0.899 (+0.043, 95% interval +0.025 to +0.063). This was our largest improvement in label agreement on the 58 expert-labelled exams. Effusion rose from 0.806 to 0.975, Baker's cyst from 0.803 to 0.979.

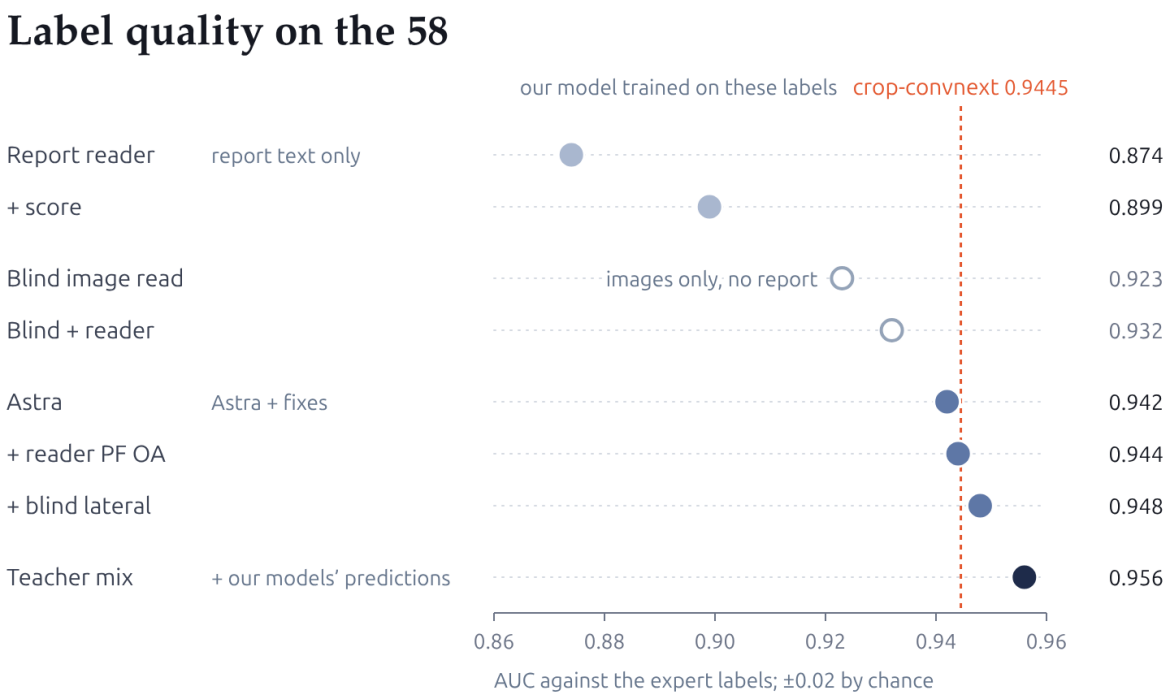
3.3 Two fixes and a second opinion

On PF OA the reader beat Astra, 0.960 against 0.930: reports describe damaged kneecap cartilage that the experts count as PF OA. So the PF OA label takes half its value from the reader. On Lateral OA a blind pass over the outer half of the knee, without the report, lifted Lateral OA from 0.856 to 0.898.

Then five teacher models trained on these labels, and their out-of-fold predictions were mixed back in. The original label keeps 0.7 where the report is the better witness (ACL, MCL, both menisci, PF OA, Lateral OA) and 0.4 elsewhere. Last came one rule from the 58: all 12 exams there that the reader called "unclear" on PF OA were positive for the experts, so in training every such exam gets at least 0.8.



One illustrative exam; each dot is one finding.



Each dot is one label version scored on the 58; the dashed line is our crop-convnext fold model.

By then the labels knew more than our models could learn, so the next gains had to come from what the models see.

3.4 Who trained on what

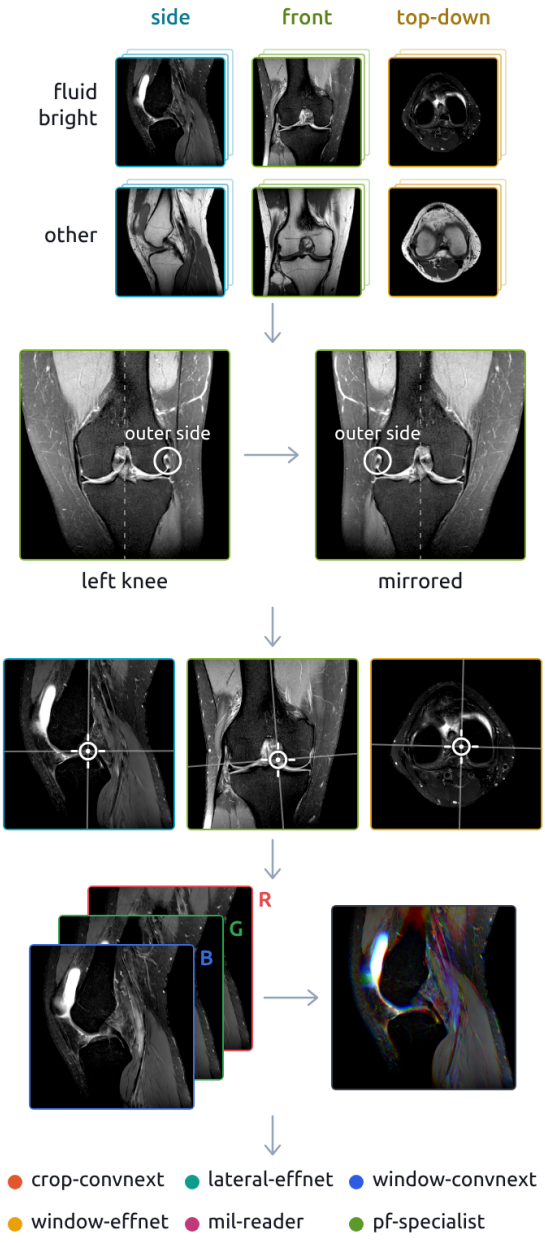
The final models train on all 4,403 usable exams, the 58 included with their expert labels. lateral-effnet stays off the 58, so it can still be checked there.

Member	Labels
crop-convnext, window-convnext	teacher mix + PF OA floor
lateral-effnet	Astra's labels with both fixes
pf-specialist	lateral-effnet's labels with a gentler PF OA floor
window-effnet	pf-specialist's labels, mixed 0.3 with mil-reader's out-of-fold predictions
mil-reader	Astra's labels

4. Inputs

Every member turns an exam into 2.5D pictures through the same steps, then cuts its own close-ups.

From exam to model input



1 Pick series

Each series: a pile of slices from one direction. A typical exam fills 5 of 6 slots.

Only clean series: 693 tilted left out. Slices sorted by position, not file order.

2 Mirror left knees

Side from slice positions: 2,245 right, 1,982 left, 118 unknown stay as is.

mil-reader without the mirror at test: left-knee lateral meniscus 0.942 → 0.879.

3 Find the knee centre

A rule from the file header (scan centre, pixel size) plus the soft-tissue middle; about 10 mm off, so boxes are generous.

A tested detector: joint 93%, kneecap 57%.

4 Stack 2.5D pictures

Thin slices averaged into 3–3.5 mm slabs; three neighbouring slices or slabs become one picture's colour channels.

5 Each member cuts its own close-ups

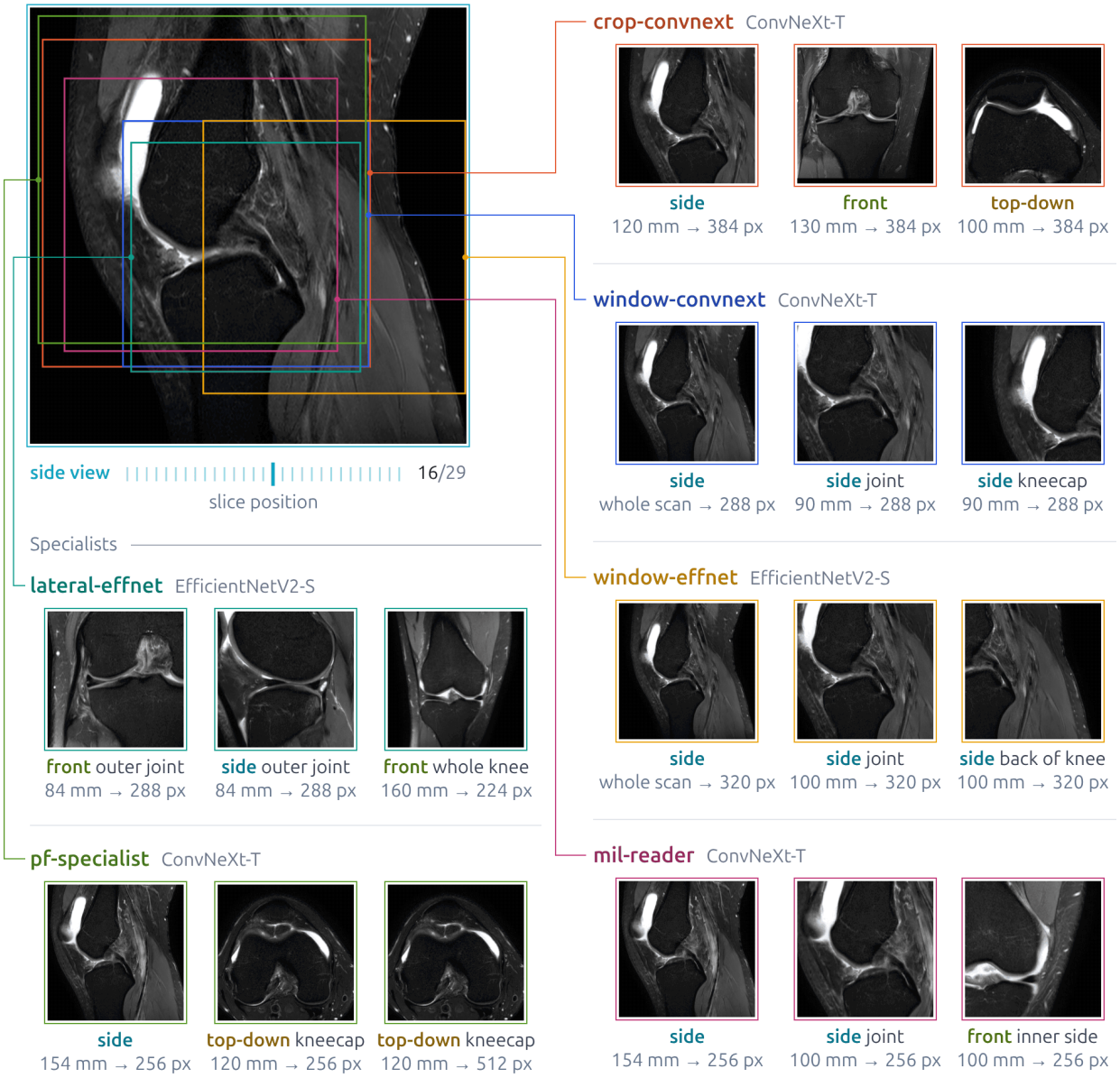
→ next figure

mil-reader and **pf-specialist** do steps 1–3 their own way.

The input steps every exam goes through, in training and at test.

mil-reader and pf-specialist came with the team merge and do the first three steps their own way. They split series by fat suppression, read the knee's side from the file tags (a small classifier trained on fastMRI decides unclear knees) and take the knee centre from the headers alone. Their close-ups follow heliosli's public recipe.

What each member sees



The scan shows where each member cuts, slice by slice; each row shows the pictures it reads.

The members differ most here, and the blend gets its value from that. One input bug cost more than most model changes: our first models read only 4 pictures per series, and reading 15 gave +0.033 on the 58. Since then every member spreads its pictures over the whole box range (mil-reader: 8 sampled slice groups per view during training, 12 evenly spaced groups at test), and at test our four average three slice offsets (0, 1, 2).

5. Models

All six encode each 2.5D picture and pool the exam per finding with attention; mil-reader and pf-specialist put a two-layer transformer over the pictures first.

5.1 The core four

- **mil-reader**, our best single member (five seeds). It grew from goodpjw2008's public 2.5D ConvNeXt-T MIL reader (Apache 2.0), which scored 0.955 public on our labels. We added the left-knee mirror, attention per finding inside every picture, and knee-centred close-ups. Its encoder was first pretrained on fastMRI knee scans with fastMRI+ boxes.
- **crop-convnext**, built from scratch by us and our simplest strong model: three joint close-ups (side, front, top-down) cut around our knee-centre rule at 384 px, three seeds.
- **window-convnext**: the window views of heliosli's public notebook (a full series view plus local views around the knee), trained our way: our labels and knee centre, every window kept, all data, three seeds. It is our best fold model on the 58 (0.9488).
- **window-effnet**, the same views on EfficientNetV2-S with frozen batch norm. Its labels lean partly on mil-reader, but it sees the knee through other views, so it disagrees with mil-reader instead of copying it.

5.2 Two specialists

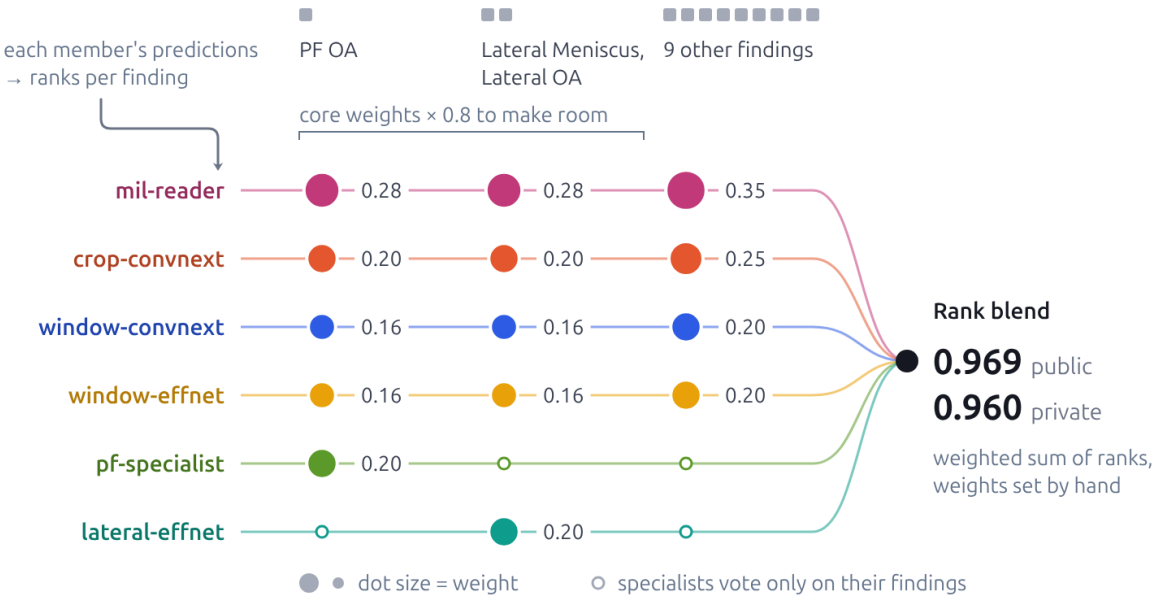
PF OA and Lateral OA were the weakest findings for every model, at about 0.92 public. So two members vote only on their own findings. To score one finding alone, we sent a probe: that finding real, the other 11 set to 0.5.

- **lateral-effnet** lifted Lateral OA to 0.932 on a public probe, against 0.920 for mil-reader. On private it fell behind (about 0.944 against 0.956), but it is the member least like the others (rank correlation about 0.92).
- **pf-specialist** is mil-reader's encoder with a second head that predicts kneecap cartilage grades, learned from MOAKS grades in the Osteoarthritis Initiative (OAI). Its PF OA probe read 0.944 public and about 0.932 private, against 0.920 and 0.908 for window-convnext. Its weight was fixed before the probe.

6. Ensemble

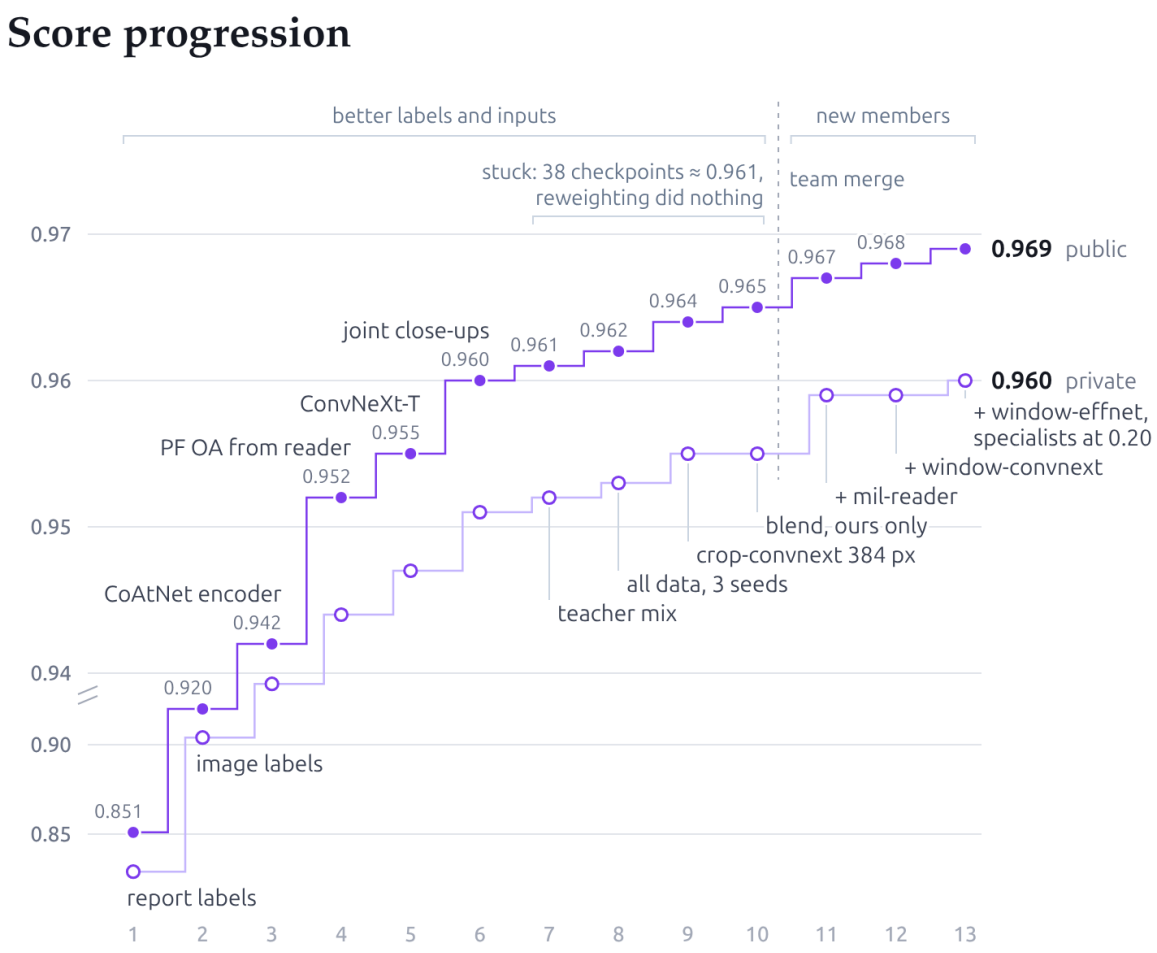
Each member's predictions become ranks per finding across the test set. The ranks are summed with the hand-set weights from the overview table, and we never searched them. On their findings the specialists take 0.20 and the core weights shrink by 0.8. The core four's submission is written first; each specialist runs within a time budget and is dropped if it fails. An unreadable exam gets a default (our members: training-set rates).

Final ensemble



Read each column top to bottom: the weight each member gets on those findings.

7. Results



Filled dots are public, rings are private; the dashed rule marks the team merge.

Better labels and inputs took our own models to 0.964 public; after that only new members helped. Public and private moved together: for submissions at 0.955 public or more, the two boards agreed on the order (rank correlation 0.977). Our chosen submission was also our best private of all 101.

8. What didn't work

- **Showing Astra the experts' answers.** It copied them, and a rule it wrote from them scored 0.80 when used blind.
- **More blind one-finding passes.** +0.066 on the 24 tuning exams, then −0.019 on the 34 held back.
- **Tuned blend weights.** They lost 0.002 to 0.007 to fixed weights on held-out exams.
- **A close-up alone.** −0.010 on Lateral Meniscus until the whole knee was added back.
- **Look-alike members.** A model at 0.959 public alone added nothing to the blend.

9. Code and acknowledgements

Training the final seeds of all six members took roughly 100 GPU hours on rented GPUs, about $100 in all, not counting research runs. Inference runs in one Kaggle notebook on T4 ×2 with internet off. Code, notebook and weights: Project Links; training settings, prompts and label builders are in the repo.

Thanks to @goodpjw2008, whose public 2.5D MIL reader was the starting point of mil-reader, and to @heliosli, whose public recipe (Apache 2.0) window-convnext and window-effnet are built on and our close-up views follow. Outside data: fastMRI knee scans with fastMRI+ boxes (pretraining and the left/right classifier of mil-reader and pf-specialist) and MOAKS cartilage grades from the Osteoarthritis Initiative (pf-specialist).

Sources

- Host ruling that commercial LLM APIs may be used: https://www.kaggle.com/competitions/rsna-knee-abnormality-detection/discussion/733965
- goodpjw2008, 2.5D ConvNeXt MIL reader (Apache 2.0): https://www.kaggle.com/code/goodpjw2008/rsna-knee-coatnet-2-5d-convnext-mil-lb-0-950
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- Osteoarthritis Initiative (https://nda.nih.gov/oai); Hunter et al., MOAKS, Osteoarthritis and Cartilage 2011.

The OAI is a public-private partnership comprised of five contracts (N01-AR-2-2258; N01-AR-2-2259; N01-AR-2-2260; N01-AR-2-2261; N01-AR-2-2262) funded by the National Institutes of Health, a branch of the Department of Health and Human Services, and conducted by the OAI Study Investigators. Private funding partners include Merck Research Laboratories; Novartis Pharmaceuticals Corporation, GlaxoSmithKline; and Pfizer, Inc. Private sector funding for the OAI is managed by the Foundation for the National Institutes of Health. This work was prepared using an OAI public use data set and does not necessarily reflect the opinions or views of the OAI investigators, the NIH, or the private funding partners.

Supplemental Q&A

> **Adel Zebiche · 2026-10-24T20:29:39.290000**
>
> Congrats on 2nd place, and thanks for such a detailed write-up, this really helps! Labelling from the images was clearly the key. Do you know roughly how much the Astra image-labelling pass cost in API fees (all ~4,400 exams at ~277 images each) ?

> **azamat1ch · 2026-10-25T12:11:08.077000**
>
> Thanks! It ran on a $200/month Codex subscription and used about two weekly allowances, roughly 400–500M tokens (mostly image input). At API prices I'd guess it would come to about $2–4k.